IEOR 221 Assignment 9

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1. The following is the result from CRR tree under different number of steps. Vega and rho are computed by central finite differences with absolute bumps of 10^{-4}.

CRR

N	NPV	Delta	Gamma	Theta	Vega	Rho
2^{11}	21.1236	0.5218	0.0060	-1.0722	67.1493	85.1271
2^{12}	21.1245	0.5217	0.0060	-1.0718	67.1536	85.1444
2^{13}	21.1249	0.5217	0.0060	-1.0716	67.1550	85.1444
2^{14}	21.1251	0.5217	0.0060	-1.0715	67.1557	85.1448
2^{15}	21.1252	0.5217	0.0060	-1.0715	67.1561	85.1452

For the accuracy of the result, we have to look into the asymptotic behavior of the CRR tree. First notice that for a payoff with enough smoothness, the price V_N computed from CRR tree using N steps has the asymptotic behavior

$$V_N = V + \frac{C}{N} + o(N^{-1}), \tag{1}$$

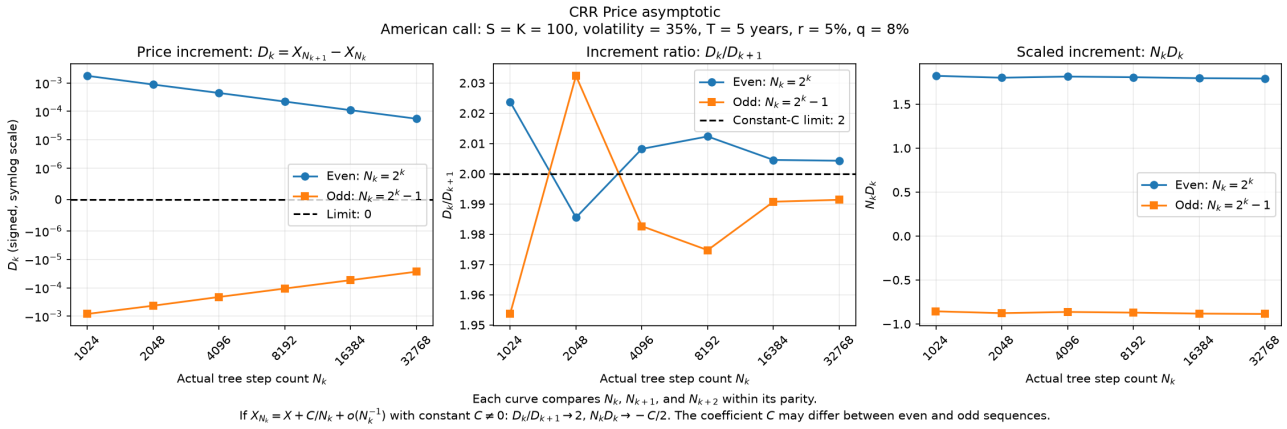
where C is a constant and V is the true price of the option. In this case, if we let

$$D_N = V_N - V = \frac{C}{N} + o(N^{-1}),$$

then

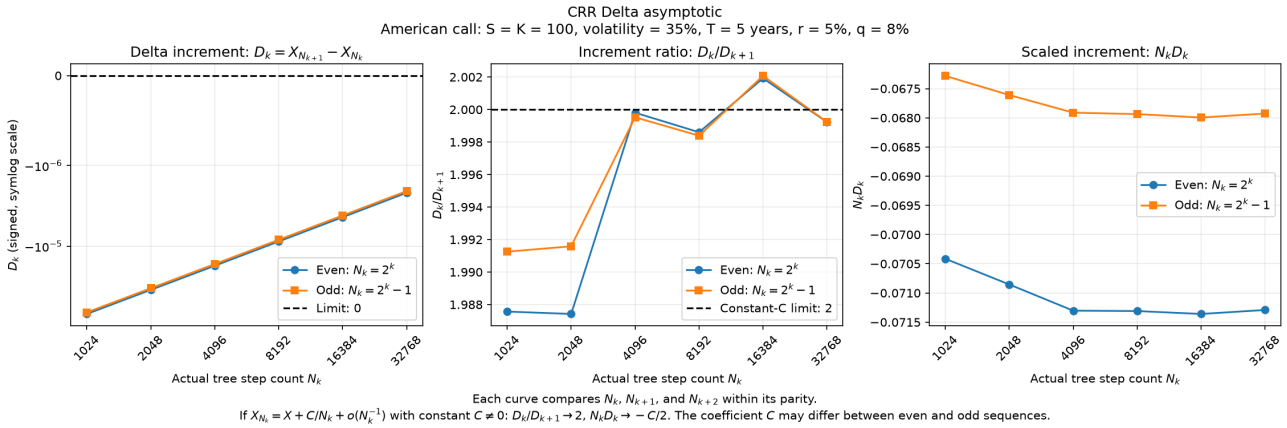
$$\lim_{N\to\infty} \frac{D_{2N}}{D_N} = 2 \qquad \lim_{N\to\infty} N \cdot D_N = -\frac{C}{2}$$

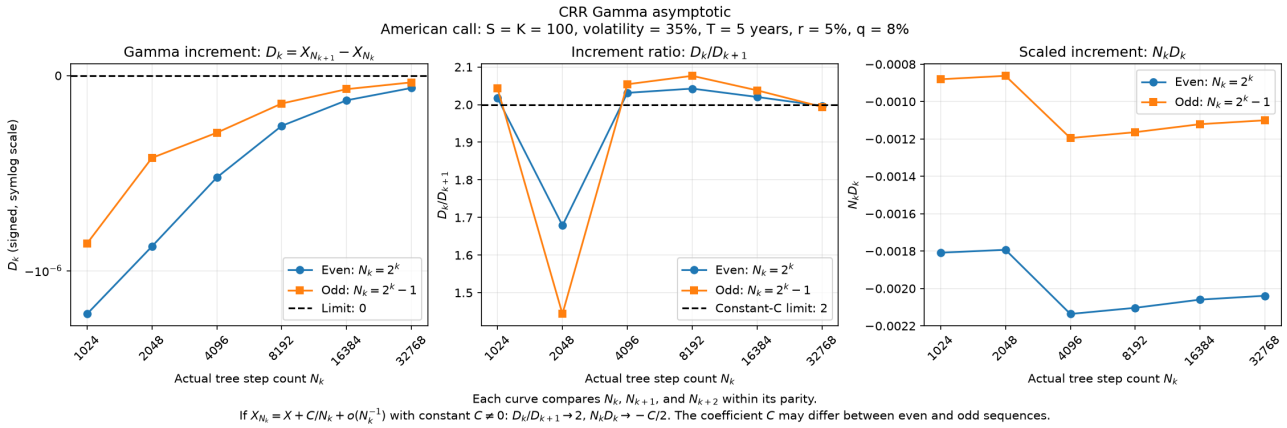
However, the payoff of the option is not smooth at the strike K, and whether K is in the ternimal price grids changes the value of C. Since the option is ATM, K is in the ternimal price grids only when N is even. So we have to separate this into cases where N is even and odd. Plotting the graph of D_{2N}/D_N (or D_{2N+1}/D_N if N is odd) and ND_N, we have the following.

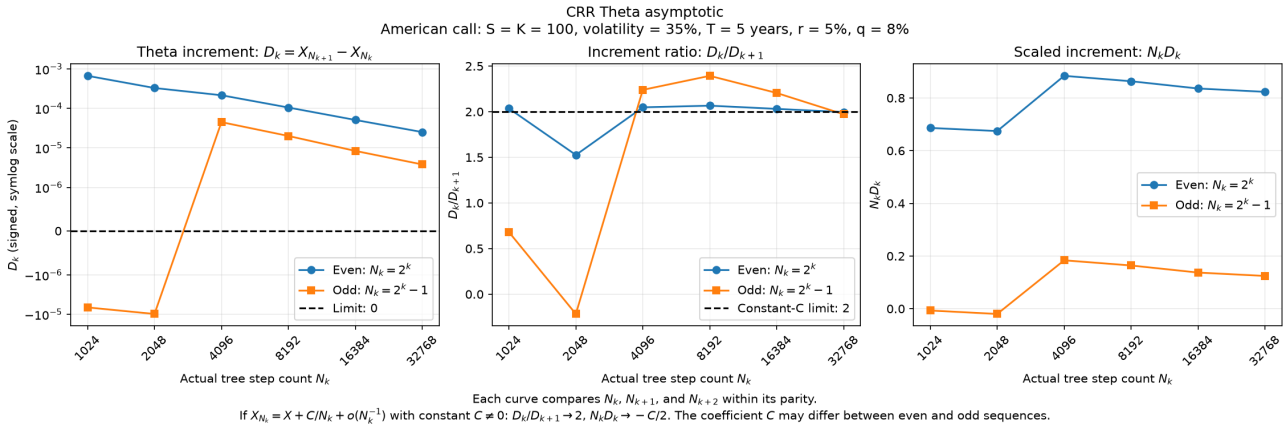


We can see that V_N does have a similar asymptotic behavior as the smooth case for even and odd N, respectively, with $C_{even} \approx -3.52$ and $C_{odd} \approx 1.74$. Hence the error of the result at 32768 steps is approximately -0.000107, and the true price should be approximately 21.1253.

For the delta, gamma, and theta, we see from the following graphs that a similar asymptotic behavior as (1) is satisfied:







Using a similar error analysis, we have delta, gamma, and theta are approximately 0.5217, 0.0060, and -1.0714, respectively.

Finally, for vega and rho, we need to take acount the effect of the bump size h, which makes

We also use a BBSR tree to give another estimation of the price and greeks, which shows a better convergance behavior:

BBSR

N	NPV	Delta	Gamma	Theta	Vega	Rho
2^{11}	21.1254	0.5217	0.0060	-1.0715	67.1584	85.1632
2^{12}	21.1254	0.5217	0.0060	-1.0714	67.1564	85.1440
2^{13}	21.1253	0.5217	0.0060	-1.0714	67.1564	85.1450
2^{14}	21.1253	0.5217	0.0060	-1.0714	67.1565	85.1456
2^{15}	21.1253	0.5217	0.0060	-1.0714	67.1565	85.1457

2. For the antithentic method, we used the opposite-sign normal samples as a pair. For the control variate method, we implemented two different control functions, which are the ternimal stock price and the price of corresponding geometric Asian call option.

Method	NPV	MC standard error
Plain MC	5.1914	0.0085
Control variate (terminal stock price)	5.1932	0.0057
Control variate (geometric Asian call)	5.1984	0.0003
Antithetic MC	5.1966	0.0069
Antithetic + control variate (terminal stock price)	5.1981	0.0049
Antithetic + control variate (geometric Asian call)	5.1987	0.0003

Each method uses 2^{20} simulated paths, the fixed seed 0, and is computed on CUDA. For antithetic methods, these paths form 2^{19} pairs. The price of the vanilla call option with the same strike and maturity is 8.8249.